

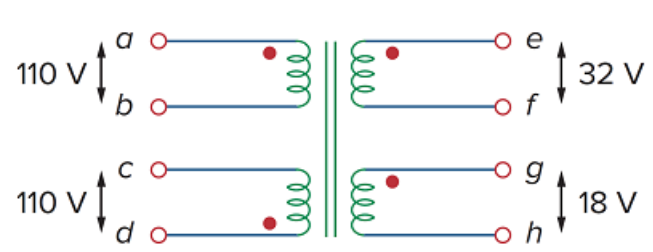
Problem

A four-winding transformer (Fig. 13.146) is often used in equipment (e.g., PCs, VCRs) that may be operated from either 110 or 220 V. This makes the equipment suitable for both domestic and foreign use. Show which connections are necessary to provide:

(a) an output of 14 V with an input of 110 V,

(b) an output of 50 V with an input of 220 V.

Figure 13.146



Step-by-step solution

Step 1 of 4

(a)

Draw the following circuit diagram:

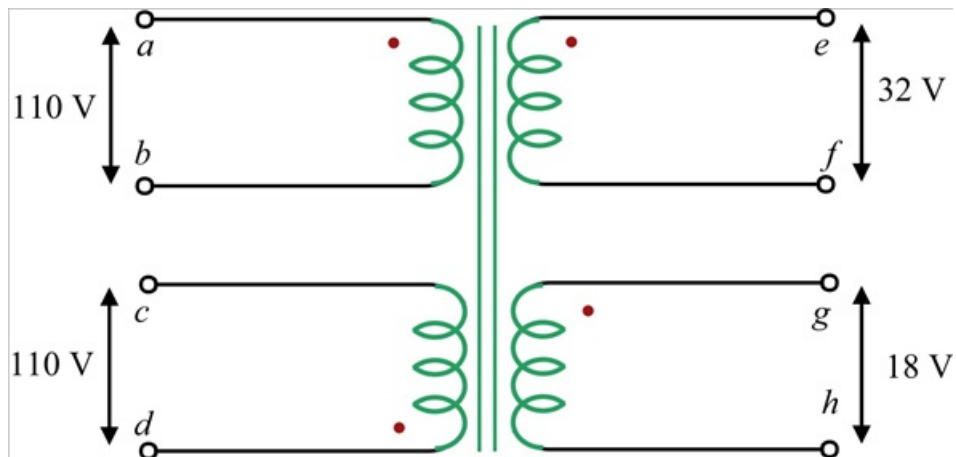


Figure 1

Step 2 of 4

Clearly from Figure 1, the output of 14 V is obtained by the subtraction of the voltages 32 V and 18 V. This is done by the series connection of terminals f to h .

The input should be 110 V. This is done by the parallel connection of terminals a to d and b to c , because they are in opposite.

Hence, draw the following circuit diagram:

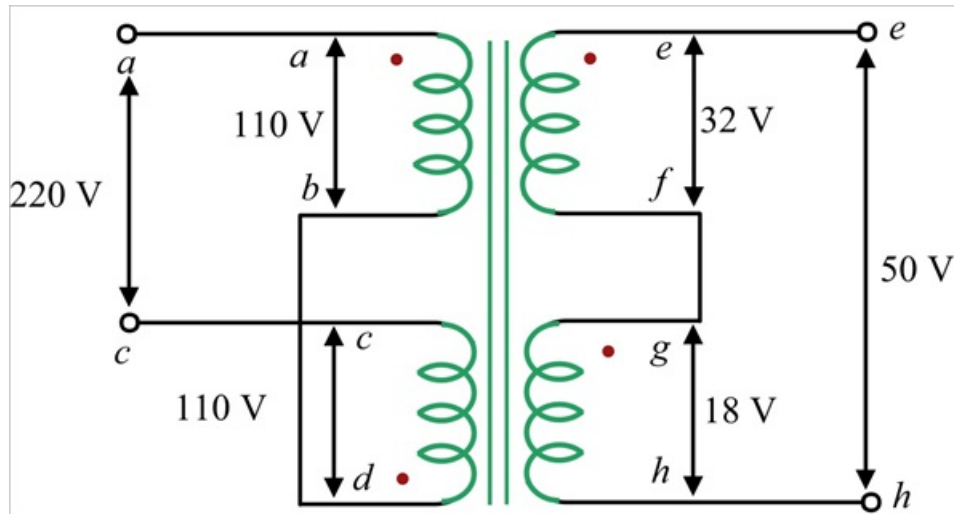


Figure 2

The necessary connections are as shown in [Figure 2](#).

Step 3 of 4

(b)

Clearly from Figure 1, the output of 50 V is obtained by the addition of the voltages 32 V and 18 V. This is done by the series connection of terminals f to g .

The input should be 220 V. This is done by the serial connection of terminals b to d because they are in opposite.

Hence, draw the following circuit diagram:

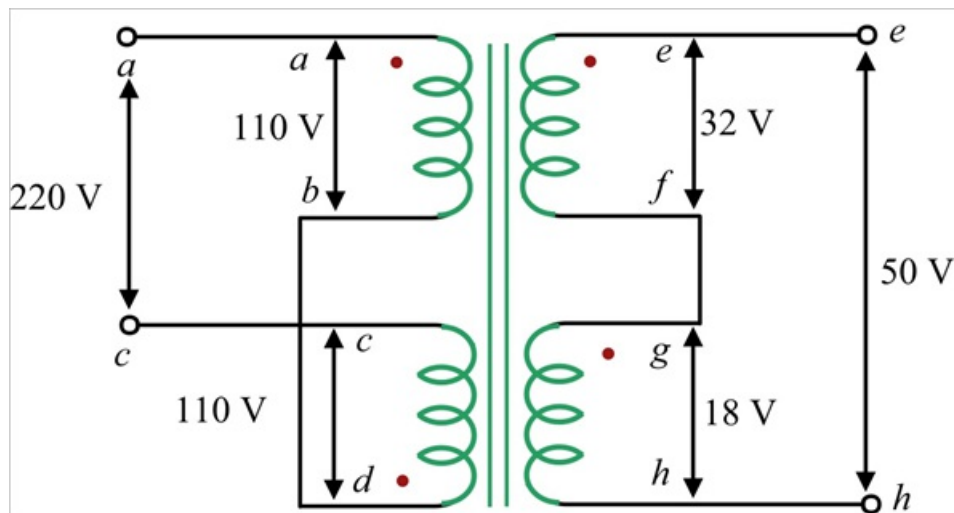


Figure 3

Step 4 of 4

The necessary connections are as shown in [Figure 3](#).